

SUBSET-CONSTRUCTION ALGORITHM

$Q^d \rightarrow$ STATI NON-MARCATI

$Q^{d+} \rightarrow$ STATI-MARCATI

$\delta^d \rightarrow$ TRANSIZIONE OR NEW STATE $\delta^d(q) = Q$

$\Sigma \rightarrow$ SIMBOLI TERMINALI (a, b, c, ...) $\emptyset \neq \epsilon \neq \emptyset$

ALGORITHM

Q^d . PUSH-BACK (ϵ -CLOSURE ($\{q_0\}$))

WHILE (Q^d IS NOT EMPTY) {

$Q = Q^d$. POP ();

Q^{d+} . INSERT (Q);

$\forall q \in \Sigma$

$T =$ GLI STATI RAGGIUNGIBILI DA Q CON q

$T' = \epsilon$ -CLOSURE (T) [COMPRESO TUTTI GLI ELEMENTI DI T]

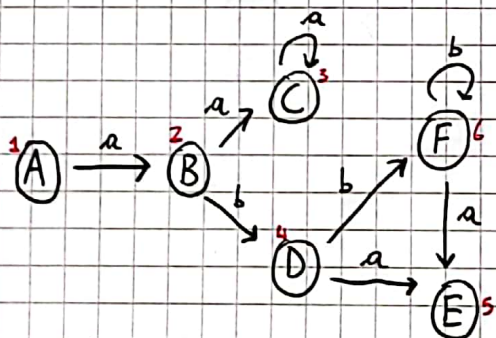
IF ($T' \neq \emptyset$ AND $T' \neq Q^d$ AND $T' \neq Q^{d+}$)

$\delta^d(Q_{NEW}) = T'$

$\delta^d(Q, q) = Q_{NEW}$

Q^d . INSERT (T')

AFD



m → NUMERAZIONE

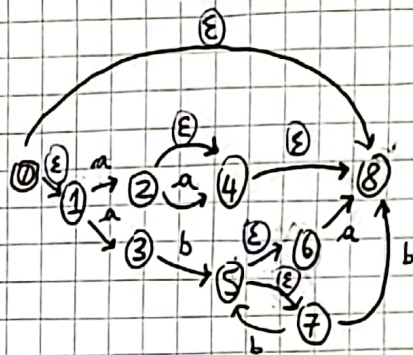
STATI SOLO

PER MIGLIORARE

EXTRA → 1 LEGGIBILITÀ.

EXAMPLE 1

AEND



ITERATION

$A = \{0, 1, 8\}$

WHILE () {

1A [$T = 2, 3$]
(a) $T' = \{2, 3, 4, 8\}$
 $\delta^d(B) = T'$

1A [X]

2B [$T = 2$]
(a) $T' = \{2, 4, 8\}$
 $\delta^d(C) = T'$

2B [$T = 5$]
(b) $T' = \{5, 6, 7\}$
 $\delta^d(D) = T'$

3C [$T = 2$]
(a) $T' = \{2, 4, 8\} \Rightarrow$

3C [X]

4D [$T = 8$]
(a) $T' = \{8\}$
 $\delta^d(E) = T'$

4D [$T = 5, 8$]
(b) $T' = \{5, 6, 7, 8\}$
 $\delta^d(F) = T'$

5E [X]

5E [X]

6F [$T = 8$]
(a) $T' = \{8\} \Rightarrow$

6F [$T = 5, 8$]
(b) $T' = \{5, 6, 7, 8\}$

Q ^d	A ^d
*	A
B	B
C	C
D	D
E	E
F	F

(A, a) = B

(B, a) = C

(B, b) = D

(C, a) = C

(D, a) = E

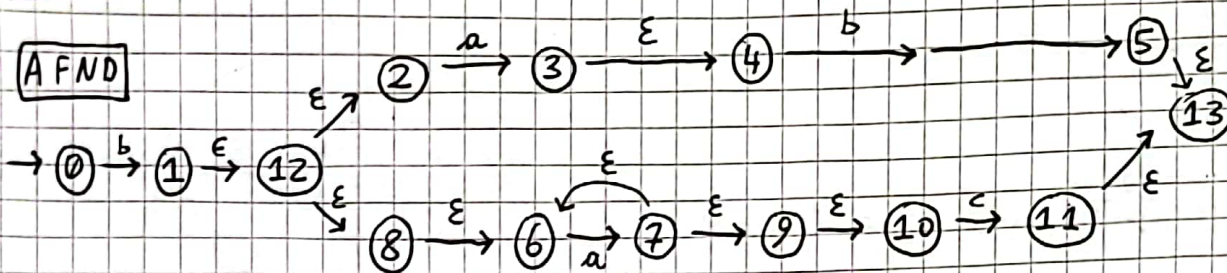
(D, b) = F

(F, a) = E

(F, b) = F

EXAMPLE 2

AFND



ITERATION

• = NEW
• = OLD

$$A = \{0\}$$

WHILE (...) {

1A [X]

$$\begin{array}{l} 1A \\ (a) \end{array} \left[\begin{array}{l} T = 1 \\ T' = \overline{1, 2, 6, 8, 9, 10} \\ \delta^A(B) = T' \end{array} \right] \quad (A, b) = B$$

$$\begin{array}{l} 5E \\ (a) \end{array} \left[\begin{array}{l} T = 7 \\ T' = \overline{6, 7, 9, 10} \end{array} \right] \quad (E, a) = E$$

5E [X]

6F [X]

1A [X]

$$\begin{array}{l} 2B \\ (a) \end{array} \left[\begin{array}{l} T = 3, 7 \\ T' = \overline{3, 4, 6, 7, 9, 10} \\ \delta^A(C) = T' \end{array} \right] \quad (B, a) = C$$

AFD

$$A = \{0\}$$

$$B = \{1, 2, 6, 8, 9, 10\}$$

$$C = \{3, 4, 6, 7, 9, 10\}$$

$$D = \{11, 13\}$$

$$E = \{6, 7, 9, 10\}$$

$$F = \{5, 13\}$$

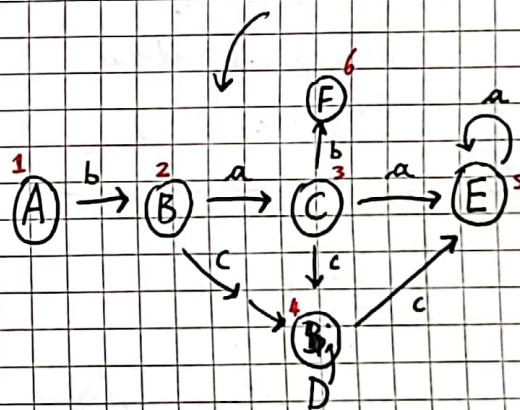
2B [X]

$$\begin{array}{l} 2B \\ (c) \end{array} \left[\begin{array}{l} T = 11 \\ T' = \overline{11, 13} \\ \delta^A(D) = T' \end{array} \right] \quad (B, c) = D$$

$$\begin{array}{l} 3C \\ (a) \end{array} \left[\begin{array}{l} T = 7 \\ T' = \overline{6, 7, 9, 10} \\ \delta^A(E) = T' \end{array} \right] \quad (C, a) = E$$

$$\begin{array}{l} 3C \\ (b) \end{array} \left[\begin{array}{l} T = 5 \\ T' = \overline{5, 13} \\ \delta^A(F) = T' \end{array} \right] \quad (C, b) = F$$

$$\begin{array}{l} 3C \\ (c) \end{array} \left[\begin{array}{l} T = 11 \\ T' = \overline{11, 13} \\ \delta^A(D) = T' \end{array} \right] \quad (C, c) = D$$



4D [X]

4D [X]

4D [X]